



Infection Prevention and Control Manual

Pertussis (Whooping cough) Policy

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Workers

Contents

Scope.....	1
Introduction	1
Risk Statement	1
Policy Principles.....	1
Responsibilities	2
1. The disease.....	2
2. Early Detection and notification.....	2
3. Case management.....	3
4. Duration of Precautions.....	4
5. Contact Management.....	4

Introduction

This policy applies to all workers but is aimed at those likely to come in contact with patients known or suspected to have pertussis (whooping cough).

Risk Statement

Non-compliance with this policy will:

Breach legislative requirements	☒	Impact on Patient Quality of Care	☐
Breach National/State/Hospital Policy	☒	Impact on Patient Safety	☒
Breach professional standards	☒	Misconduct	☐
Breach SCGOPHCG Mission & Values	☐	Work Safety	☒
Other:	☐		

Policy Principles

This policy provides detailed information on the infection prevention and management (IPM) of patients with known or suspected pertussis and workers exposed to the bacterium *Bordetella pertussis*. It is designed to ensure the risk to patients, workers and visitors is minimised.

Responsibilities

All workers must adhere to this policy.

Pertussis infection is a **notifiable disease** under the *Public Health Act 2016*.¹ The medical team responsible for the patient must notify the [Communicable Disease Control Directorate \(CDCD\), Health Department of WA](#) on confirmation of the disease.

1. The disease

Pertussis is caused by the bacterium *Bordetella pertussis*, a gram-negative bacillus. It causes a prolonged coughing illness characterised by an initial catarrhal phase followed by a paroxysmal coughing phase. Complications may include pneumonia.

Bordetella parapertussis is a closely related bacteria that does not produce the pertussis toxin and causes a milder disease. In the absence of an outbreak, droplet precautions are not required for patients with *B.parapertussis* infection.

1.1 Mode of transmission

Pertussis is transmitted by large droplets or direct contact with respiratory secretions from infectious cases.

1.2 Clinical presentation

The initial catarrhal phase may include symptoms such as rhinorrhoea, fever and a mild cough. The paroxysmal coughing phase is distinguished by a prolonged, characteristic “whooping” cough, episodes of cyanosis and may include post-tussive vomiting. Adult patients may not present with classic symptoms.

1.3 Incubation period

The incubation period is variable, commonly 7 to 10 days (range from 4 to 21 days).

1.4 Infectious period

Patients with pertussis are infectious until 5 days after receiving appropriate systemic antibiotic treatment, 21 days after the onset of cough or 14 days after the onset of paroxysmal cough. (refer to section 4).

1.5 Diagnosis

A laboratory diagnosis should be sought for all clinically suspected pertussis cases as it is a notifiable disease.

Nucleic acid amplification testing (eg PCR) is the test of choice in the first 3 weeks of illness.

An upper respiratory tract specimen (eg nose/throat swab) should be collected with a dry swab and sent to the laboratory for testing with a request for ‘pertussis PCR’. The sensitivity of PCR testing is highest early in the illness and may be a less reliable test in the third and fourth weeks of illness. If the presentation is delayed, serological testing should be considered but the interpretation of these results will need to consider vaccination history.

Contact a Clinical Microbiologist to discuss testing if required.

2. Early Detection and notification

2.1 Notification during business hours, 0800-1600hrs Monday-Friday, excluding public holidays

Notify IPM Unit of any suspected or confirmed pertussis infection in any patient. The IPM unit will liaise with the on-call Clinical Microbiologist and Health Safety and Wellbeing, via:

SCGH: Extension 34555 or email InfectionControl.SCGH@health.wa.gov.au any time.

OPH: Extension 78099 or email InfectionControl.OPH@health.wa.gov.au any time.

2.2 Notification after hours

Notify the on-call Clinical Microbiologist of any suspected or confirmed pertussis infection in an

inpatient. The on-call Clinical Microbiologist can be contact via switchboard during the following hours:

- Monday – Friday 1600-2100hrs
- Weekends and public holidays 0800-2100hrs.

3. Case management

Until the patient can be managed in a single room with droplet precautions, staff are to provide the patient with a surgical mask and as appropriate assist the patient with the correct donning and doffing of the mask. The patient is to be prioritised for inpatient admission to a single room with droplet precautions. If single room is not immediately available manage patient in an area with spatial separation from other patients (>2m) and where possible create a physical barrier between other patients e.g., privacy screen and use a portable air purifier. Place a green DROPLET PRECAUTION sign outside the patient's room for the duration of the infectious period. The reason for isolation should be explained to the patient and family.

Non-essential workers must not enter the room, e.g. medical, nursing and allied health students.

Refer to [SCGOPHCG IP&M Manual, Standard and Transmission based precautions Policy](#) for further information.

3.1 Clinical management

Clinical teams should consider the use of antibiotics to reduce the period of communicability. Refer to the [Australian Therapeutic Guidelines](#) - Pertussis or [Australian Centre for Disease Control](#) – Pertussis guideline for more information if required.

3.2 Workers

Pregnant or immunosuppressed workers should not care for the patient. All healthcare workers are encouraged to be up-to date with their vaccination for pertussis.

3.3 Hand hygiene

Perform hand hygiene in accordance with the '5 moments for hand hygiene'. Use hand wash solution or alcohol-based hand rub.

3.4 Personal protective equipment

PPE to be worn as for droplet precautions. At a minimum, a surgical face mask and eye protection must be donned prior to entry of the patient room.

3.5 Equipment

- keep equipment in room to a minimum
- supplies and equipment are to be disposable where possible
- dedicate non-critical items to the room, e.g. stethoscope, tourniquet
- patient charts should be left outside the room
- reusable equipment is to be cleaned with combined detergent/disinfectant wipe and allowed to dry before use on another patient.

3.6 Routine and discharge cleaning

Cleaning staff must wear PPE as per droplet precautions.

Follow cleaning guidelines for isolation room.

3.7 Linen

Standard precautions apply. Linen that is wet or soiled with body substances must be contained in a plastic bag prior to disposal in the linen bag.

3.1 Eating utensils

Standard precautions apply. Disposable crockery and cutlery are not required.

3.2 Transport to other departments e.g. radiology

During the infectious period:

- this should be kept to a minimum
- the receiving department must be informed prior to transfer
- the patient must wear a surgical face mask
- ensure no visitors or other workers are in the lift during transfer
- patients on oxygen therapy should be changed to nasal prongs and wear a surgical face mask if possible
- Workers accompanying the patient do not routinely need to wear PPE. PPE (as per droplet precautions) is only required if the patient is unable to wear a surgical face mask
- hand hygiene must be performed after transporting the patient.

3.3 Transport of laboratory specimens

Follow standard precautions.

3.4 Visitors

All visitors must wear a surgical facemask and protective eyewear when visiting the patient for the duration of the infectious period. Visitors should be instructed on how to perform hand hygiene and on the correct donning and doffing of PPE.

Visitors who are pregnant, immunocompromised or <6 months of age should not visit.

Ensure visitors are not present when undertaking aerosol generating procedures on the patient.

4. Duration of Precautions

Droplet precautions should continue until the case is no longer infectious which is defined as:

- Completion of 5 days of an appropriate antibiotic, or
- 21 days after the onset of any cough

5. Contact Management

Identifying contacts of Pertussis is undertaken to determine those patients and workers who may have had exposure, to assess their risk, and as appropriate, provide advice and implement post-exposure prophylaxis. During routine business hours (Monday – Friday, 0800-1600 hrs, not including public holidays), identification of staff contacts is undertaken, the Health, Safety and Wellbeing (HSW) Department. Patient contacts will be identified by the IPM Department.

Contact tracing and management of **non-hospital contacts** will be coordinated by the Western Australian Communicable Disease Control Directorate (CDCD) and the relevant Public Health Unit.³ If a patient has proven pertussis, the treating clinician should inform CDCD by completing an [Infectious and Related Diseases Notification Form](#).

Close contacts are defined as:

- People who have had face-to-face exposure (within 1 metre), to an infectious case for a single period of one hour³
- People who have occupied the same room for ≥12 hours
- Family and household members (living and sleeping in the same household or dormitory).

All contacts should receive a [Department of Health Pertussis information sheet](#).

High risk close contacts, as defined in Table 1, may require prophylaxis with antibiotics (chemoprophylaxis).

Table 1: High risk close contacts**High risk close contact may be defined as:**

- Household contacts who are expectant parents (or carers) in the last month of pregnancy
- All household members where there is an infant <6 months of age present
- Healthcare staff working in a maternity ward or newborn nursery
- Childcare staff who look after infants <6 months of age
- Children who have close contact in childcare with other children <6 months of age
- Pregnant people in the last month of pregnancy
- Contacts at risk of severe disease (shorter durations of exposure may be considered)
- This may include patients with pre-existing asthma, COPD or morbid obesity.

Chemoprophylaxis regimens are available in the Australian Therapeutic Guidelines⁴ (available on the SCGH Intranet). A pertussis disease [fact sheet](#), available from the Healthy WA website, should be provided to contacts.

Active immunisation of contacts does not ameliorate the risk of developing pertussis. Vaccination of contacts should be offered as per the [Australian Immunisation Handbook](#).

5.1 Management of hospital patient contacts

Chemoprophylaxis should be considered if a patient is a high-risk contact as defined above.

The IPM Unit work in collaboration with Clinical Microbiologist to identify high risk patient contacts, including exposed patient who may have been discharged. The Clinical Microbiologist will liaise with the treating team to discuss recommendations for chemoprophylaxis if indicated. Treating teams will be responsible for coordinating treatments, including for discharged patients.

Close contacts who remain an in-patient following exposure, regardless of receiving chemoprophylaxis, should be managed with standard precautions, and monitored for symptoms. Place symptomatic close contacts on droplet precautions until testing supports an alternative diagnosis.

5.2 Management of worker contacts

Contacts are defined as workers who have had face-to-face exposure (within 1 metre) to an infectious case, for a single period of at least one hour without appropriate PPE (as for droplet precautions).

The wearing of a surgical face mask is considered protective against respiratory droplets. Exposure of the eyes to respiratory droplets is not considered an indication for chemoprophylaxis.

When indicated, chemoprophylaxis should be administered as soon as possible after exposure. Chemoprophylaxis administered more than 14 days after exposure is unlikely to be of benefit³.

Vaccination should be offered to all health care workers who are incompletely vaccinated^{3,8}.

5.2.1 Management during working hours (8:00-15:00 Monday to Friday excluding public holidays)

When a confirmed case of pertussis is identified, HSW will be informed by the IPM Unit or Clinical Microbiologist. HSW will:

- Liaise with the relevant wards to compile a list of HCW contacts and obtain details of the exposure.
- Determine, in discussion with the Clinical Microbiologist, if chemoprophylaxis is indicated for individual HCWs based on details of exposure

- Counsel the HCW about risks and benefits of chemoprophylaxis
- Prescribe chemoprophylaxis where relevant (HCWs may decline chemoprophylaxis)
- Provide the HCW with a [fact sheet](#) available from the Healthy WA website.

5.2.2 After-hours management

Outside working hours, the on-call Clinical Microbiologist will inform the After-hours Nurse Manager (AHNM) for SCGH (Ext. 73188) or OPH (Ext. 78098 or pager 113) of the suspected/confirmed case. The AHNM will liaise with the relevant wards to compile a list of HCW contacts and obtain details of exposure. Based on this information, the on-call Clinical Microbiologist will determine if chemoprophylaxis is indicated for individual HCWs. Medical Officers on the relevant wards (e.g. ED, ICU) may be asked to assess and counsel the HCW and prescribe chemoprophylaxis under the direction of the on-call clinical Microbiologist. All queries regarding HCW exposure and management should be directed to the AHNM or to HSW the next working day. The list of HCW contacts completed by the AHNM should be emailed or faxed to HSW.

5.2.3 Potential Worker contacts who are no longer at work

In some cases, HCW contacts potentially requiring chemoprophylaxis will no longer be at work. These HCWs should be contacted by HSW or their supervisor, and details of their exposure obtained. If chemoprophylaxis is indicated, the HCW will be advised to attend the HSW clinic or their GP, depending on the time of day/week. Rarely, the HCW may be advised to attend the Emergency Department.

5.2.4 Worker contacts with Pertussis exposure in the community

Workers who have been identified as a close contact following exposure to a person with confirmed pertussis in the community will be managed by Public Health.

Related documents

SCGOPHCG Infection Prevention and Management Manual: Standard and Transmission-Based Precautions Policy

References

1. *Public Health Act 2016 (WA)*. https://www.legislation.wa.gov.au/legislation/statutes.nsf/main_mrtitle_13791_homepage.html
2. Heymann, D.L. *Control of Communicable Diseases Manual*. 19th ed. American Public Health Association. 2008. P 455-461.
3. Communicable Disease Network Australia (CDNA) Series of National Guidelines. Pertussis: National Guidelines for Public Health Units. 2024. [cited 2024 Nov 20] Available from: <https://www.health.gov.au/resources/publications/pertussis-whooping-cough-cdna-national-guidelines-for-public-health-units?language=en>
4. Pertussis [published 2019 Apr]. In: *Antibiotic - Therapeutic Guidelines*. Melbourne: Therapeutic Guidelines Limited; accessed {20 Nov 2024}. <https://www.tg.org.au>
5. Centres for Disease Control. Whooping cough (Pertussis). 2024 [cited 2024 Nov 20] Available from: <https://www.cdc.gov/pertussis/site.html#hpc>
6. Australian Technical Advisory Group on Immunisation (ATAGI). *Australian Immunisation Handbook*, Australian Government Department of Health and Aged Care, Canberra, 2022, immunisationhandbook.health.gov.au
7. Calderón TA, Coffin SE, Sammons JS. Preventing the Spread of Pertussis in Pediatric Healthcare Settings. *J Pediatric Infect Dis Soc*. 2015 Sep;4(3):252-9. doi: 10.1093/jpids/piu056. Epub 2014 Jun 17. PMID: 26407429
8. Cornia P, Lipsky BA. Pertussis Infection: Epidemiology, microbiology and pathogenesis. In: UpToDate, Calderwood SB, Kaplan SL, Hussain Z, White N (Ed), Wolters Kluwer. (Accessed on December 1,2024).

Authorisation & Version Control

Policy Sponsor	Medical Co-Director, Specialty and Ambulatory Services				
Policy Contact	Co-ordinator of Nursing, Infection Prevention and Management				
Date First Issued:	31/03/2025	Last Reviewed:	-	Review Date:	31/03/2028
Version No.	1				
Endorsed by:	SCGOPHCG Infection Prevention and Management Committee			Date:	31/03/2025
Standards Applicable:	National Standards (version 2)   Standard 3: Preventing and managing healthcare associated infections				

For full details regarding, development, review, consultation, approval, endorsement - contact the Policy Officer for the submission form

This document can be made available in alternative formats on request for a person with a disability.